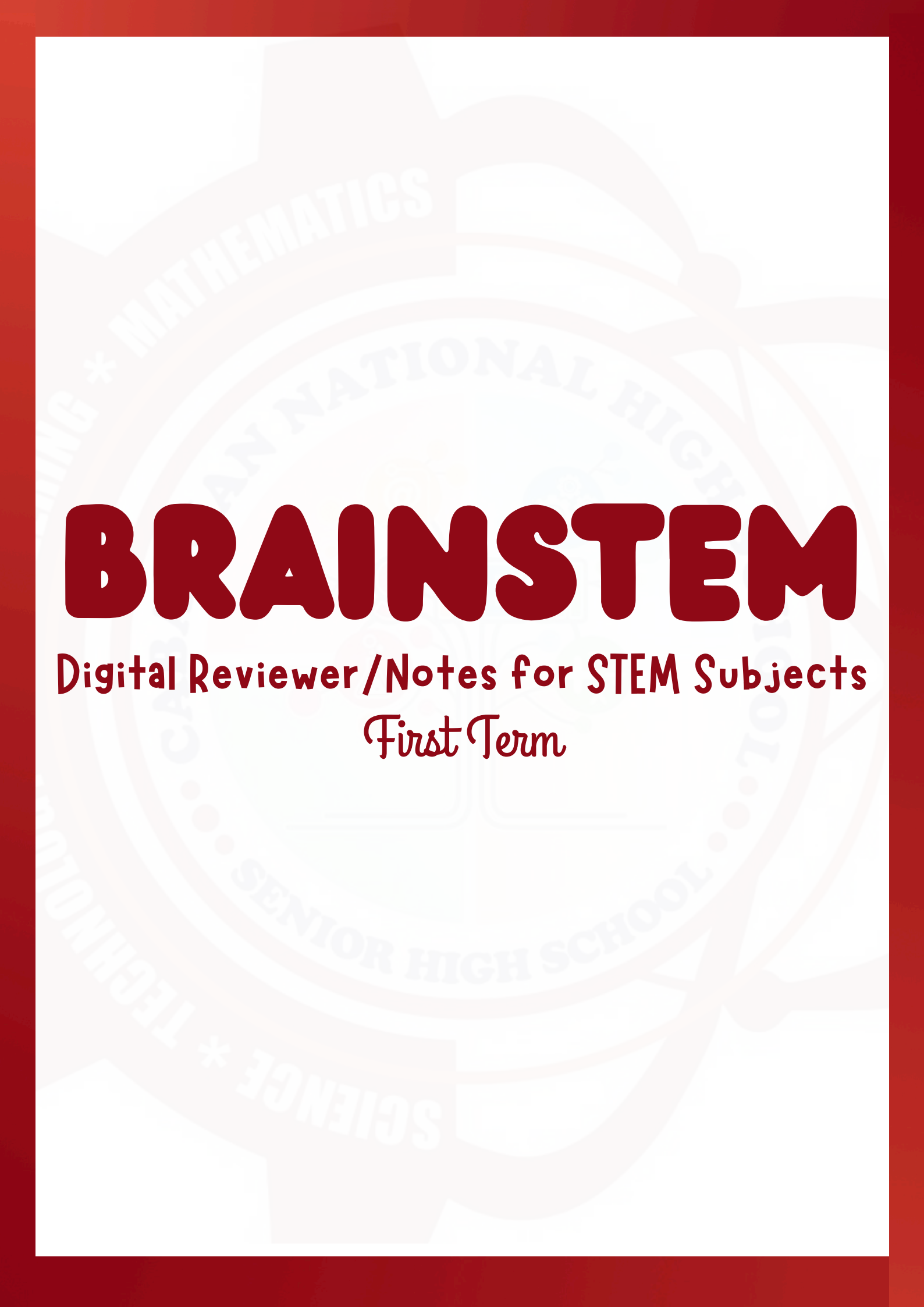




# BRAINSTEM

Digital Reviewer/Notes for STEM Subjects  
*First Term*

# PHYSICS IN THE FIELD OF ENGINEERING



## Measurement and Vectors

### 1. Measurement

- is the comparison of an unknown quantity with a known quantity.
- Direct measurement - with the use of a measuring device.
- ex: Fundamental quantities.
- Indirect Measurement - through calculating using known equation of formula.
- ex: Device quantities.
- Non Standard unit of measurement

### Hand Span

- distance between the tip of the thumb and the tip of the little finger of fully stretched hand.

### Ubit

- distance between the tip of the middle finger (outstretched) and the elbow.

### Fathorn

- is the length of the outstretched arms

### Ancient Instruments

- Footsteps
- Rope
- Hand span
- Arm length

### Ancient Instruments

- Ruler
- Measuring tape
- Screw gauge
- Vernier caliper

## Fundamental units and quantities

| Quantity         | Unit     | Symbol | Definition                                                                                                                                             |
|------------------|----------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length           | meter    | m      | • distance travelled by light in a vacuum in $\frac{1}{299,792,458}$ sec.                                                                              |
| Mass             | kilogram | kg     | • actual mass of platinum iridium cylinder kept of international bureau of weights and measures since 1889.                                            |
| Time             | second   | sec.   | • refers to the duration for 9,192,631,790 cycles of the microwave radiation corresponding to the transition between two lowest states of cesium atom. |
| Temperature      | kelvin   | k      | • it means fraction of $\frac{1}{273.16}$ of the triple point of the water.                                                                            |
| Electric current | ampere   | A      | • this is the constant                                                                                                                                 |

| System | Length     | Mass     | Time |
|--------|------------|----------|------|
| FPS    | Foot       | Pound    | sec. |
| CGS    | centimeter | gram     | sec. |
| MKS    | meter      | kilogram | sec. |

## Metric english equivalent

| Metric        | English           |
|---------------|-------------------|
| (54) 2.54 cm  | 1 inch (in)       |
| 1 meter       | 39.37 inches (in) |
| 1 km          | 0.62 miles        |
| 1 Liter       | 1.06 quartz (qt)  |
| 250 ml        | 1 cup (c)         |
| 1 kg          | 2.2 pounds        |
| 92.28g) 28.3g | 1 ounce           |
| 1 m           | 3.28 feet         |
| 90 cm         | 1 yard            |
| 3.97 L        | 1 gallon          |

# PHYSICS IN THE FIELD OF ENGINEERING



## Rules for counting significant digits

- All non-zero digits and any zeros contained between non-zero digit count  
30042 = 6 significant digits
- Leading zeros don't count  
0.000034 = 2 significant digits
- Trailing zeros count if there is a decimal point  
0.0002500 = 4 significant
- Trailing zeros may or may not count if there is no decimal point. so we go with the most conservative answer.  
190000 = 2 significant figures

| Accuracy                                                                                                                                                                                                                                                                                                                                    | Precision                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>How close the data or results are to the goal</li> <li>Measures of quantity to reality</li> <li>Takes into account the true "true value"</li> <li>"Bad results" would be far from the goal</li> <li>Can be determined with a single measurement</li> <li>Does not speak about the quality</li> </ul> | <ul style="list-style-type: none"> <li>How close the data or results are to each other</li> <li>Measures how well measurements can be reproduced.</li> <li>Doesn't take into account the accepted value.</li> <li>bad results would be scattered</li> <li>Need several measurements to be determined</li> <li>Speak about the quality.</li> </ul> |

| Systematic error                                                                                                       | Random error                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Occurs on a result of a flaw in the experimental design or apparatus</li> </ul> | <ul style="list-style-type: none"> <li>Cused by unpredictable changes in the experiment current in each of two long parallel wires. 1.0 m apart 2 x 10<sup>-7</sup> newton per meter length of each wire.</li> </ul> |

|     | Length     | Mass     | Time |
|-----|------------|----------|------|
| FPS | Foot       | Pound    | sec. |
| CGS | centimeter | gram     | sec. |
| MKS | meter      | kilogram | sec. |

|                     |         |     |
|---------------------|---------|-----|
| Luminous intensity  | Candela | cd  |
| Amount of substance | mole    | mol |

- Derived quantities and units
  - combination of two or more basic quantities
  - ex. speed, velocity, acceleration, force, weight.

## System of units

- British / english system
  - was first adopted in England and later became widely used in the US.
  - also known as the FPS system (pounds)
- Metric system
  - originated in France in 1791, has the units of meter, centimeter kilogram and second.

## Temperature conversion formulas

- Celsius to Kelvin :  $K = C + 273.15$
- Kelvin to Celsius :  $C = K - 273.15$
- Fahrenheit to Celsius:  $C = (F - 32) \times 5/9$
- Celsius to Fahrenheit:  $F = C (5/9) + 32$
- Fahrenheit to Kelvin:  $K = (F - 32) \times 5/9 + 273.15$
- Kelvin to Fahrenheit:  $F = (K - 273.15) \times 9/5 + 32$

## Conversion of units

- From large to small unit
  - to convert from a large to small unit, multiply the given number of the large unit by the number of small units contained in one large
- Small - Large
  - divide the given number of the small (to large unit) by the number of small units contained in one large unit.

## Scalar vs Vector

- Scalar quantity has magnitude only.
- Vector quantity has magnitude and direction.
- Scalar quantities are described by a number and unit only.
- Vector quantities are described by a number, unit and direction.
- Examples of scalar quantities: distance, speed, mass, time, temperature.
- Examples of vector quantities: displacement, velocity acceleration force.

# PHYSICS IN THE FIELD OF ENGINEERING



## Costumary Measurement

| Lenght                                                | Weight                                | Capacity                                                                         |
|-------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------|
| 1 ft = 12 inch<br>1 yard = 3ft<br>1 mile = 1760 yards | 1 pound = 16 ounces<br>1 ton = 2, 000 | 1 cup = 8 fluid oz<br>1 pint = 2 cups<br>1 quart = 2 pint<br>1 gallon = 4 quartz |

## Metric Measurement

| Lenght                                       | Weight                          | Capacity               |
|----------------------------------------------|---------------------------------|------------------------|
| 1 km = 100 m<br>1 m = 100 cm<br>1 cm = 10 mm | 1 kg = 100 g<br>1 gram = 100 mg | 1 l = 1000 milliliters |

## Mass

1 ton = 1000 kg  
 1 kilogram = 2.2 pounds (lb)  
                   = 0.0685 slug  
 1 lb = 453.6 gm = 0.031 slug  
 1 slug = 32.174 lb = 14. 59 kg  
 1 lb = 16.02  
 1 oz = 28.35g  
 1 troy oz = 31.1234768 g

## Percentage error

- Relative error shown as a percentage

Formula:

Absolute error = V true - V observed

Realtive error = V true - V observed/v true

Percentage error = V true - v observed/v true x 100

- are errors caused due to the error in the instrument and usually can be corrected by simple calculation of improved experiment tenchniques

### Common sources of systematic errors

- Faulty calibration of measuring instruments
- Poorly maintained instruments
- Faulty reading of instruments by the user

### Absolute error

- Difference between the actual and measured value

- are errors of observation which measurement is just as likely to be larger or smaller than the true or accepted value

### Common sources of random errors

- Problems estimating a quantity that lies between the graduations (the lines) on an instrument
- The inability to read an instrument.

### Relative error

- The absolute error divided by the actual measurement.
- Relative error:  
Absolute error/  
measured value.